# Collectively Big, Locally Small: Online Supplement

Systematic search and coded corpus

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This supplement accompanies the survey *Collectively Big, Locally Small*. Section S1 documents the systematic database search summarized in Section 5.4 of the paper: sources, query strings, screening rules and their verification, the recall of the search against the coded corpus, the validation of the dictionary coder, and the field-level map. Section S2 lists the 290 coded works. The simulation study of the Graph Hierarchical Borrowing blueprint summarized in Section 8.7, its revision and the extensions tried are reported in full in the companion paper [Simões et al. 2026] and its supplement. Every number in this supplement is generated by the scripts released with the paper (`prisma/`) and read into the text as LaTeX macros.

## S1 SYSTEMATIC SEARCH

### S1.1 Sources and query strings

The search was run on 8 October 2026 for records published from 1 January 2010 to the search date. PubMed was queried through the NCBI E-utilities with every term restricted to title and abstract (`[tiab]`). medRxiv and the arXiv records indexed by Europe PMC were queried through the Europe PMC REST API (source `PPR`, field `TITLE_ABS`). Europe PMC indexes only the life-science categories of arXiv. A direct search of the arXiv API was attempted but abandoned because the API rejected the full query; computer-science preprints are therefore covered only when they were also published in an indexed venue. Every query string is stored verbatim in `prisma/data/search_log.json` together with the hit counts; the PubMed string was

```
("small data"[tiab] OR "small sample*"[tiab] OR "limited data"[tiab] OR "data scarcity"[tiab] OR
"few-shot"[tiab] OR "low-resource"[tiab] OR "rare disease*"[tiab] OR "label-efficient"[tiab] OR
"data-efficient"[tiab] OR federated[tiab] OR decentralized[tiab] OR decentralised[tiab] OR
"privacy-preserving"[tiab]) AND (health*[tiab] OR clinical[tiab] OR medic*[tiab] OR
hospital*[tiab] OR patient*[tiab]) AND ("machine learning"[tiab] OR "deep learning"[tiab] OR
"transfer learning"[tiab] OR "synthetic data"[tiab] OR "data augmentation"[tiab] OR "active
learning"[tiab] OR "semi-supervised"[tiab] OR "self-supervised"[tiab] OR "foundation
model*"[tiab] OR "hierarchical model*"[tiab]) AND 2010/01/01:2026/10/08[dp]
```

Scopus, Web of Science, IEEE Xplore and the ACM Digital Library were not searched, because they require a subscription or an API key. Exports from them (RIS, BibTeX or CSV) can be added with `prisma/import_exports.py`; they are then deduplicated and screened with the other records, and the flow diagram and every number are regenerated.

### S1.2 Screening rules

Records were deduplicated by DOI and by normalized title, keeping the PubMed version when a preprint had also been published. Titles and abstracts were then screened by the rules of Table S1, applied in order. The rules operationalize criteria I1, I2 and X1 to X3 of Section 5: they keep every record in which a scarcity term of the query refers to scarce, fragmented or poorly labeled data and a learning method is present, and they remove the frequent uses of the query terms that do not refer to data scarcity.

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Table S1. Screening rules, applied in order; the first rule that applies decides.

| Rule | Excludes records whose | *n* |
|---|---|---|
| A1 | abstract is missing, or language is not English | 46 |
| A2 | publication type is editorial, letter, comment, erratum, guideline, consensus statement, protocol, audio-visual material or historical article, or which are retracted (X1) | 54 |
| R1 | only data-scarcity term refers to a resource-limited setting ("low-resource settings, countries, hospitals") | 421 |
| R2 | only data-scarcity term is "small sample" in a sentence on limitations or future work | 264 |
| R3 | only decentralization term refers to decentralized clinical trials or care delivery | 14 |
| R4 | title is a reporting guideline, checklist, statement, position paper, policy or consensus (X1) | 8 |
| R5 | title and abstract contain no learning method | 72 |
| | Eligible on title and abstract | 5,204 |

### S1.3 Verification by a single reviewer

A single reviewer judged two random samples of 100 records each against criteria I1, I2 and X1 to X3, reading title and abstract; each sample drew 60 or 40 records from the included side and as many from the excluded side of the rules then in use. The first sample led to the removal of one rule, which excluded records whose scarcity term appears only in the author keywords (4 of its 8 false exclusions came from it). A rule added after the first sample, which excluded "limited data" in the sense of a gap in clinical knowledge, removed small-training-set studies in 3 of the 7 sampled records it excluded in the second sample and was also removed. Against all 200 judgments, the final rules have a precision of 63% and a negative predictive value of 94%; the sensitivity within the samples is 94%, but the samples are balanced by design, so this is not the sensitivity in the full set. A stricter rule, which required the scarcity term in the title or in a sentence that also mentions data, labels, training or a model, raised precision to 73% but lowered sensitivity to 77% and was not adopted. The judgments are released in `prisma/data/verify_judgments.json`.

### S1.4 Recall against the coded corpus

A search is usually validated by its relative recall against a reference set of known relevant studies. Our reference set is the coded corpus: 201 of its works were published since 2010 and evaluated on health data, 144 of these are indexed in the searched sources (found in Europe PMC by DOI or title), and the protocol query retrieved 31 of them (22%). To test whether a better query would have found them, we split the 140 PubMed-indexed works at random into two halves and evaluated broadened variants of the query on PubMed (Table S2). Adding 39 terms to the scarcity block (sample size, multi-site, external validation, generalizability, shift, privacy, annotation and related terms) and 16 terms to the method block (artificial intelligence, neural networks, prediction models and related terms) multiplied the number of records by thirteen and raised recall to about one half. The literature on learning from scarce health data is not well delimited by vocabulary. A large part of it reports a method or a failure without naming scarcity at all, which is why the paper's synthesis rests on the snowballed corpus.

### S1.5 Dictionary coder and field-level map

Each eligible record was coded from its title and abstract by dictionaries for the six provenance channels, the ten scarcity classes and the data modality (`prisma/classify.py`). The primary channel is the channel whose terms match most often, with title matches counted twice. On the 147 corpus works whose abstracts are available in Europe PMC, the dictionary coder agreed with the manual primary channel in 72% of cases (Cohen's $\kappa = 0.65$) and recovered 63% of the manually

Table S2. Number of PubMed records and relative recall on two random halves of the PubMed-indexed corpus works, for the protocol query and for broadened variants.

| Query | PubMed records | Recall, first half | Recall, second half |
|---|---|---|---|
| Protocol query (Section 5.4) | 5,885 | 17% | 27% |
| Scarcity block broadened | 40,584 | 39% | 44% |
| Method block broadened | 11,178 | 21% | 31% |
| Both broadened | 75,983 | 51% | 54% |

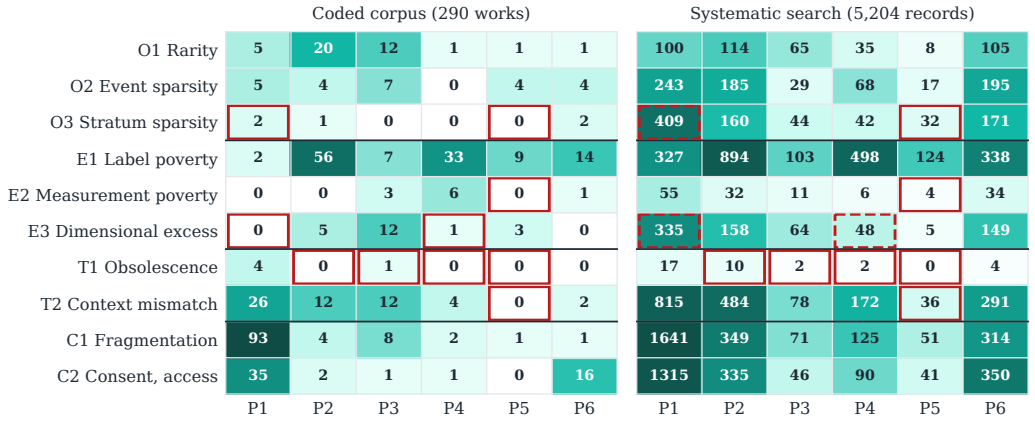



Fig. S1. Left: works of the coded corpus targeting each scarcity class through each channel, as in Figure 9 of the paper; red outlines mark the ten gap cells. Right: records of the systematic search coded by dictionary, shaded by percentile rank among the 60 cells. Solid outlines: gaps confirmed (bottom third of the field-level map); dashed outlines: gaps not confirmed.

assigned scarcity classes. Figure S1 sets the coverage of the coded corpus (Figure 9 of the paper) next to the field-level map. Because the dictionaries for stratum sparsity (which includes the term bias) and for dimensional excess (which includes omics terms) are broad, their rows are inflated in the field-level map; Section 5.4 of the paper discusses the three gap cells that the search does not confirm.

## S2 THE CODED CORPUS

Table S3 lists all 290 works of the corpus with their coding under the scheme of Table 3 of the paper. The same data are released as a CSV file, together with the scripts that compute every count in Sections 5 and 7 and draw Figures 8 and 9.

Table S3. The coded corpus (290 works), grouped by primary provenance channel and sorted by year. Sec.: secondary channel. Scarcity: classes of Figure 3 the work targets or documents. Type: Method, Theory, Study (single-dataset empirical study), Multi-site (training or validation across institutions), Resource (dataset, benchmark or infrastructure), Review, Evidence (documented failure or risk). General: method not specific to health data.

| Work | Modality | Sec. | Scarcity | Type |
|---|---|---|---|---|
| *P1 Pooling* (102) | | | | |
| DerSimonian and Laird [1986] | Tabular | P3 | C1, T2 | Method |

| Work | Modality | Sec. | Scarcity | Type |
|---|---|---|---|---|
| Sweeney [2002] | Tabular | | C2 | Method |
| Xiao and Boyd [2004] | General | | C1 | Theory |
| Boyd et al. [2006] | General | | C1 | Theory |
| Dwork et al. [2006] | General | | C2 | Theory |
| Gentry [2009] | General | | C2 | Theory |
| Nedić and Ozdaglar [2009] | General | | C1 | Theory |
| Riley et al. [2010] | Tabular | | C1, O2 | Review |
| Overhage et al. [2012] | EHR | | C2 | Resource |
| Wu et al. [2012] | Tabular | | C1 | Method |
| Jiang et al. [2013] | Tabular | | C1 | Resource |
| Gaye et al. [2014] | Tabular | | C1, C2 | Resource |
| Hripcsak et al. [2015] | EHR | | C1, C2 | Resource |
| Lu et al. [2015] | Tabular | | C1 | Resource |
| Abadi et al. [2016] | General | | C2 | Method |
| Jochems et al. [2016] | Tabular | | C1 | Multi-site |
| Johnson et al. [2016] | EHR | | C1, C2 | Resource |
| Mandel et al. [2016] | EHR | | C2 | Resource |
| Blanchard et al. [2017] | General | | C1 | Method |
| Bonawitz et al. [2017] | General | | C1, C2 | Method |
| Jochems et al. [2017] | Tabular | | C1 | Multi-site |
| Kim et al. [2017] | EHR | | C1 | Method |
| Lian et al. [2017] | General | | C1 | Theory |
| McMahan et al. [2017] | General | | C1 | Method |
| Mironov [2017] | General | | C2 | Theory |
| Papernot et al. [2017] | General | P4 | C2 | Method |
| Smith et al. [2017] | General | P3 | C1, T2 | Method |
| Vanhaesebrouck et al. [2017] | General | P3 | C1, T2 | Method |
| Bellet et al. [2018] | General | P3 | C1, T2 | Method |
| Brisimi et al. [2018] | EHR | | C1 | Method |
| Chang et al. [2018] | Imaging | | C1 | Multi-site |
| Cho et al. [2018] | Omics | | C1, C2 | Method |
| Gupta and Raskar [2018] | General | | C1 | Method |
| Lee et al. [2018] | EHR | | C1, C2 | Method |
| McMahan et al. [2018] | General | | C1, C2 | Method |
| Reps et al. [2018] | EHR | | C2 | Resource |
| Vepakomma et al. [2018] | Multimodal | | C1, C2 | Method |
| Zhao et al. [2018] | General | | C1, T2 | Method |
| Choudhury et al. [2019] | EHR | | C1, C2 | Method |
| Huang et al. [2019] | EHR | | C1, T2 | Multi-site |
| Jordan et al. [2019] | General | | C1 | Theory |
| Li et al. [2019] | Imaging | | C1, C2 | Method |
| Mohri et al. [2019] | General | | C1, O3 | Method |
| Poirot et al. [2019] | Imaging | | C1, C2 | Method |
| Raisaro et al. [2019] | Multimodal | | C1, C2 | Method |
| Roy et al. [2019] | Imaging | | C1 | Method |
| Sheller et al. [2019] | Imaging | | C1 | Method |
| Suchard et al. [2019] | EHR | | C1, C2, O2 | Multi-site |
| Yang et al. [2019] | General | | C1 | Review |
| Beyan et al. [2020] | Multimodal | | C1, C2 | Resource |
| Chen et al. [2020b] | Signals | P2 | C1, T2 | Method |
| Deist et al. [2020] | Tabular | | C1 | Multi-site |
| T. Dinh et al. [2020] | General | P3 | C1, T2 | Method |
| Duan et al. [2020b] | EHR | | C1, O2 | Method |
| Duan et al. [2020a] | EHR | | C1, O2 | Method |
| Fallah et al. [2020] | General | P2 | C1, T2 | Method |
| Ghosh et al. [2020] | General | | C1, T2 | Method |
| Hsieh et al. [2020] | General | | C1, T2 | Method |
| Kaissis et al. [2020] | Imaging | | C1, C2 | Review |
| Karimireddy et al. [2020] | General | | C1, T2 | Method |
| Koloskova et al. [2020] | General | | C1 | Theory |
| Li et al. [2020c] | General | | C1, T2 | Method |
| Li et al. [2020a] | Imaging | P2 | C1, T2 | Multi-site |
| Lin et al. [2020] | General | P4 | C1, T2 | Method |
| Rieke et al. [2020] | Multimodal | | C1, C2 | Review |
| Roth et al. [2020] | Imaging | | C1, T2 | Multi-site |
| Sheller et al. [2020] | Imaging | | C1, O1 | Multi-site |
| Wei et al. [2020] | General | | C1, C2 | Method |
| Bai et al. [2021] | Imaging | | C1, T1 | Multi-site |
| Collins et al. [2021] | General | P2 | C1, T2 | Method |
| Dayan et al. [2021] | Multimodal | | C1, T1, O2 | Multi-site |
| Dou et al. [2021] | Imaging | | C1, T1 | Multi-site |
| Froelicher et al. [2021] | Omics | | C1, C2 | Method |
| Guo et al. [2021] | Imaging | | C1 | Method |

| Work | Modality | Sec. | Scarcity | Type |
|---|---|---|---|---|
| Huang et al. [2021a] | General | | C1, T2 | Method |
| Kairouz et al. [2021] | General | | C1 | Review |
| Kaissis et al. [2021] | Imaging | | C1, C2 | Method |
| Li et al. [2021a] | General | P3 | C1, T2 | Method |
| Li et al. [2021b] | General | | C1, T2 | Method |
| Liu et al. [2021] | Imaging | P6 | C1, T2 | Method |
| Marfoq et al. [2021] | General | | C1, T2 | Method |
| Sarma et al. [2021] | Imaging | | C1 | Multi-site |
| Tramèr and Boneh [2021] | General | P2 | C2 | Method |
| Vaid et al. [2021] | EHR | | C1, T1 | Multi-site |
| Warnat-Herresthal et al. [2021] | Multimodal | | C1, O3 | Multi-site |
| Xu et al. [2021] | Multimodal | | C1 | Review |
| Ziller et al. [2021] | Imaging | | C2 | Method |
| Adnan et al. [2022] | Imaging | | C1, C2 | Method |
| Cyffers and Bellet [2022] | General | | C1, C2 | Theory |
| Jiang et al. [2022] | Imaging | | C1, T2 | Method |
| Lu et al. [2022] | Imaging | | C1, E1 | Method |
| Nguyen et al. [2022] | Multimodal | | C1 | Review |
| Ogier du Terrail et al. [2022] | Multimodal | | C1, T2 | Resource |
| Pati et al. [2022] | Imaging | | C1, O1 | Multi-site |
| Saldanha et al. [2022] | Imaging | | C1 | Multi-site |
| Tedeschini et al. [2022] | Imaging | | C1 | Method |
| Xu et al. [2022] | Imaging | | C1, T2 | Method |
| Kalra et al. [2023] | Imaging | | C1, C2 | Method |
| Lu et al. [2023] | General | | C1 | Method |
| Ogier du Terrail et al. [2023] | Imaging | | C1, O1 | Multi-site |
| Tan et al. [2023] | General | | C1, T2 | Review |
| Yan et al. [2023] | Imaging | P4 | C1, E1 | Method |
| *P2 Proxy* (60) | | | | |
| Baxter [2000] | General | | O1, E1 | Theory |
| Ben-David et al. [2010] | General | | T2 | Theory |
| Pan and Yang [2010] | General | | E1, T2 | Review |
| Choi et al. [2016] | EHR | | E1, T2 | Method |
| Gulshan et al. [2016] | Imaging | | E1 | Study |
| Maurer et al. [2016] | General | | O1 | Theory |
| Shin et al. [2016] | Imaging | | E1 | Study |
| Tajbakhsh et al. [2016] | Imaging | | E1 | Study |
| Esteva et al. [2017] | Imaging | | E1 | Study |
| Finn et al. [2017] | General | | O1, E1 | Method |
| Kamnitsas et al. [2017] | Imaging | P4 | T2 | Method |
| Snell et al. [2017] | General | | O1, E1 | Method |
| Alsentzer et al. [2019] | Text | | E1 | Method |
| Chen et al. [2019] | Imaging | | E1 | Method |
| Gurovich et al. [2019] | Imaging | | O1 | Method |
| Prabhu et al. [2019] | Imaging | | O1, O3 | Method |
| Raghu et al. [2019] | Imaging | | E1 | Study |
| Zhang et al. [2019] | EHR | | O1, E1 | Method |
| Brown et al. [2020] | General | | E1 | Method |
| Lee et al. [2020] | Text | | E1 | Method |
| Li et al. [2020b] | EHR | | E1, E3 | Method |
| Mahajan et al. [2020] | Imaging | | O1 | Method |
| Roy et al. [2020] | Imaging | | E1, O1 | Method |
| Tripuraneni et al. [2020] | General | | E1 | Theory |
| Wang et al. [2020] | General | | O1, E1 | Review |
| Azizi et al. [2021] | Imaging | P4 | E1, T2 | Method |
| Du et al. [2021] | General | | E1 | Theory |
| Huang et al. [2021b] | Imaging | P4 | E1 | Method |
| Radford et al. [2021] | General | | E1 | Method |
| Rasmy et al. [2021] | EHR | | E1, O2 | Method |
| Sowrirajan et al. [2021] | Imaging | P4 | E1 | Method |
| Steinberg et al. [2021] | EHR | | E1 | Method |
| Zhou et al. [2021] | Imaging | P4 | E1 | Method |
| Agrawal et al. [2022] | Text | | E1 | Method |
| Guan and Liu [2022] | Imaging | | T2 | Review |
| Hospedales et al. [2022] | General | | O1, E1 | Review |
| Hsieh et al. [2022] | Imaging | | O1 | Method |
| Hu et al. [2022] | General | | E1 | Method |
| Mei et al. [2022] | Imaging | | E1 | Resource |
| Tiu et al. [2022] | Imaging | P4 | E1 | Method |
| Wang et al. [2022] | Imaging | P4 | E1 | Method |
| Zhang et al. [2022] | Imaging | P4 | E1 | Method |
| Banerjee et al. [2023] | Multimodal | P3 | O1 | Review |

| Work | Modality | Sec. | Scarcity | Type |
|---|---|---|---|---|
| Hegselmann et al. [2023] | Tabular | P3 | E1, O2 | Method |
| Hollmann et al. [2023] | Tabular | P3 | E3, O2 | Method |
| Jiang et al. [2023] | Text | | E1 | Method |
| Moor et al. [2023] | Multimodal | | E1, O1 | Review |
| Nori et al. [2023] | Text | | E1 | Study |
| Singhal et al. [2023] | Text | | E1 | Method |
| Theodoris et al. [2023] | Omics | | E1, E3, O1 | Method |
| Wornow et al. [2023] | EHR | | E1, T2 | Review |
| Zhou et al. [2023] | Imaging | P4 | E1, T2 | Method |
| Chen et al. [2024] | Imaging | | E1, O1 | Method |
| Cui et al. [2024] | Omics | | E1, E3 | Method |
| Lu et al. [2024] | Imaging | | E1, O1 | Method |
| Ma et al. [2024] | Imaging | | E1 | Method |
| Pai et al. [2024] | Imaging | | E1, O1 | Method |
| Vorontsov et al. [2024] | Imaging | | E1, O1 | Method |
| Xu et al. [2024] | Imaging | | E1 | Method |
| Hollmann et al. [2025] | Tabular | P3 | E3, O2 | Method |
| *P3 Prior* (25) | | | | |
| Stein [1956] | General | P1 | O1, C1 | Theory |
| Efron and Morris [1973] | General | P1 | O1, C1 | Theory |
| Tibshirani [1996] | General | | E3 | Method |
| Steyerberg et al. [2000] | Tabular | | O2, E3 | Study |
| Lake et al. [2015] | General | | O1 | Method |
| Choi et al. [2017a] | EHR | | E3, O1 | Method |
| Rotmensch et al. [2017] | EHR | P4 | E3 | Method |
| Ma et al. [2018] | EHR | | E3, O1 | Method |
| Arjovsky et al. [2019] | General | | T1, T2 | Method |
| Raissi et al. [2019] | General | | O1, E3 | Method |
| Shang et al. [2019] | EHR | | E3 | Method |
| Subbaswamy et al. [2019] | EHR | | T2 | Method |
| Castro et al. [2020] | Imaging | | T2, E2 | Review |
| Kissas et al. [2020] | Imaging | | E1 | Method |
| Lagergren et al. [2020] | Tabular | | E1, E3 | Method |
| Sahli Costabal et al. [2020] | Signals | | O1, E1 | Method |
| Van Calster et al. [2020] | Tabular | | O2 | Study |
| Gulrajani and Lopez-Paz [2021] | General | | T2 | Study |
| Karniadakis et al. [2021] | General | | O1, E3 | Review |
| Köhler et al. [2021] | Text | | O1 | Resource |
| Lu et al. [2021a] | Tabular | | O1 | Method |
| Riley et al. [2021] | Tabular | | O2 | Study |
| Schölkopf et al. [2021] | General | | T2 | Review |
| Zhang et al. [2021] | Multimodal | | T2 | Study |
| von Rueden et al. [2023] | General | | O1, E3 | Review |
| *P4 Unlabeled* (23) | | | | |
| Schölkopf et al. [2012] | General | | E1 | Theory |
| Agarwal et al. [2016] | EHR | | E1, E2 | Method |
| Halpern et al. [2016] | EHR | P3 | E1, E2 | Method |
| Bai et al. [2017] | Imaging | | E1 | Method |
| Ratner et al. [2017] | General | P3 | E1, E2 | Method |
| Wang et al. [2017] | Imaging | | E1, E2 | Resource |
| Ilse et al. [2018] | Imaging | | E1 | Method |
| Campanella et al. [2019] | Imaging | | E1 | Method |
| Cheplygina et al. [2019] | Imaging | P2 | E1 | Review |
| Fries et al. [2019] | Imaging | | E1 | Method |
| Irvin et al. [2019] | Imaging | | E1, E2 | Resource |
| Wang et al. [2019] | Text | P3 | E1 | Method |
| Yu et al. [2019] | Imaging | | E1 | Method |
| Chaitanya et al. [2020] | Imaging | P2 | E1 | Method |
| Chen et al. [2020a] | General | P2 | E1 | Method |
| Ouyang et al. [2020] | Imaging | P2 | E1, O1 | Method |
| Sohn et al. [2020] | General | | E1 | Method |
| Tajbakhsh et al. [2020] | Imaging | P6 | E1, E2 | Review |
| Taleb et al. [2020] | Imaging | P2 | E1 | Method |
| Li et al. [2021c] | Imaging | | E1 | Method |
| Lu et al. [2021b] | Imaging | | E1 | Method |
| He et al. [2022] | General | P2 | E1 | Method |
| Krishnan et al. [2022] | Multimodal | P2 | E1 | Review |
| *P5 Procurement* (15) | | | | |
| Peduzzi et al. [1996] | Tabular | | O2 | Theory |
| Settles [2009] | General | | E1 | Review |
| Figueroa et al. [2012] | Text | | E1 | Method |

| Work | Modality | Sec. | Scarcity | Type |
|---|---|---|---|---|
| Cho et al. [2015] | Imaging | | E1 | Study |
| Gal et al. [2017] | General | | E1 | Method |
| Yang et al. [2017] | Imaging | | E1 | Method |
| Smailagic et al. [2018] | Imaging | | E1 | Method |
| Balki et al. [2019] | Imaging | | E1, O1 | Review |
| Ghorbani and Zou [2019] | General | | C1 | Method |
| Riley et al. [2019b] | Tabular | | O2, E3 | Theory |
| Riley et al. [2019a] | Tabular | | E3 | Theory |
| van Smeden et al. [2019] | Tabular | P3 | O2 | Theory |
| Riley et al. [2020] | Tabular | | O2, E3 | Theory |
| Budd et al. [2021] | Imaging | | E1 | Review |
| Nath et al. [2021] | Imaging | | E1 | Method |
| *P6 Re-expression* (32) | | | | |
| Chawla et al. [2002] | Tabular | | O2 | Method |
| Goodfellow et al. [2014] | General | | E1 | Method |
| Choi et al. [2017b] | EHR | | C2 | Method |
| Bowles et al. [2018] | Imaging | | E1 | Method |
| Fernández et al. [2018] | Tabular | | O2 | Review |
| Frid-Adar et al. [2018] | Imaging | | E1, O2 | Study |
| Shin et al. [2018] | Imaging | | E1, C2 | Method |
| Walonoski et al. [2018] | EHR | P3 | C2 | Resource |
| Zhang et al. [2018] | General | | E1 | Method |
| Baowaly et al. [2019] | EHR | | C2 | Method |
| Beaulieu-Jones et al. [2019] | Tabular | P1 | C2 | Method |
| Jordon et al. [2019] | Tabular | P1 | C2 | Method |
| Shorten and Khoshgoftaar [2019] | General | | E1 | Review |
| Xu et al. [2019] | Tabular | | C2 | Method |
| Yi et al. [2019] | Imaging | | E1 | Review |
| Zhao et al. [2019] | Imaging | | E1 | Method |
| Goncalves et al. [2020] | EHR | | C2 | Study |
| Ho et al. [2020] | General | | E1 | Method |
| Yoon et al. [2020] | EHR | | C2 | Method |
| Zhang et al. [2020] | EHR | | C2 | Method |
| Chaitanya et al. [2021] | Imaging | P4 | E1 | Method |
| Chen et al. [2021a] | Multimodal | | C2, O1 | Review |
| Chlap et al. [2021] | Imaging | | E1 | Review |
| Jordon et al. [2022] | General | | C2 | Review |
| Stadler et al. [2022] | Tabular | | C2 | Study |
| van den Goorbergh et al. [2022] | Tabular | | O2 | Study |
| Borisov et al. [2023] | Tabular | P2 | C2 | Method |
| Kazerouni et al. [2023] | Imaging | | E1 | Review |
| Khader et al. [2023] | Imaging | | E1, C2 | Method |
| Kotelnikov et al. [2023] | Tabular | | C2 | Method |
| Ktena et al. [2024] | Imaging | | O3, T2 | Method |
| Shumailov et al. [2024] | General | | O3 | Evidence |
| *No remedy: documented scarcity, failure or risk* (33) | | | | |
| El Emam et al. [2011] | Tabular | | C2 | Review |
| Button et al. [2013] | General | | O2 | Evidence |
| Shokri et al. [2017] | General | | C2 | Evidence |
| Varoquaux [2018] | Imaging | | O1, O2 | Evidence |
| Zech et al. [2018] | Imaging | | T2 | Evidence |
| Bagdasaryan et al. [2019] | General | | O3, C2 | Evidence |
| Brasil et al. [2019] | Multimodal | | O1 | Review |
| Christodoulou et al. [2019] | Tabular | | O2 | Review |
| Jayaraman and Evans [2019] | General | | C2 | Evidence |
| Kim et al. [2019] | Imaging | | T2 | Review |
| Liu et al. [2019] | Imaging | | T2 | Review |
| Obermeyer et al. [2019] | EHR | | O3 | Evidence |
| Rocher et al. [2019] | Tabular | | C2 | Evidence |
| Schwarz et al. [2019] | Imaging | | C2 | Evidence |
| Zhu et al. [2019] | General | | C1, C2 | Evidence |
| Geiping et al. [2020] | General | | C1, C2 | Evidence |
| Nagendran et al. [2020] | Imaging | | T2 | Review |
| Nguengang Wakap et al. [2020] | Tabular | | O1 | Evidence |
| Oakden-Rayner et al. [2020] | Imaging | | O3 | Evidence |
| Schaefer et al. [2020] | Multimodal | | O1 | Review |
| Willemink et al. [2020] | Imaging | | E1, E2, C2 | Review |
| Wynants et al. [2020] | Multimodal | | O2, T1 | Evidence |
| Chen et al. [2021b] | Multimodal | | O3 | Review |
| Finlayson et al. [2021] | Multimodal | | T1, T2 | Evidence |
| Koh et al. [2021] | Multimodal | | T2 | Resource |
| Roberts et al. [2021] | Imaging | | O2, T1 | Evidence |

| Work | Modality | Sec. | Scarcity | Type |
|---|---|---|---|---|
| Wong et al. [2021] | EHR | | T2 | Evidence |
| Carlini et al. [2022] | General | | C2 | Evidence |
| Gichoya et al. [2022] | Imaging | | O3, T2 | Evidence |
| Packhäuser et al. [2022] | Imaging | | C2 | Evidence |
| Varoquaux and Cheplygina [2022] | Imaging | | O1, T2 | Evidence |
| Yu et al. [2022] | Imaging | | T2 | Review |
| Hatamizadeh et al. [2023] | Imaging | | C1, C2 | Evidence |

## S3 DATA AND REPORTED RESULTS OF EVERY CODED WORK

For each of the 206 corpus works evaluated on health data, this section gives the data it uses and its headline result (Section 7.2 of the paper). They were extracted from the work's Europe PMC abstract when available (`prisma/data/corpus_lookup.jsonl`) and otherwise from its arXiv abstract, its PubMed record or its own page. Every non-empty field is supported by a verbatim quote, stored with its source in `data/corpus_extraction.json`; for works taken from their Europe PMC abstract, `scripts/corpus_extract.py` checks that each quote occurs in the abstract or title. Where an abstract does not name its data, the entry describes the data in the abstract's words or reads "not stated"; "no number in abstract" means the abstract gives no headline number, not that the paper has none. Sizes and results are quoted as the authors state them. Works are grouped by primary channel and ordered by year.

**P1 Pooling (62 works)**

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [DerSimonian and Laird 1986] | 1986 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Sweeney 2002] | 2002 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Riley et al. 2010] | 2010 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Overhage et al. 2012] | 2012 | EHR | 10 observational healthcare databases | no number in abstract |
| [Wu et al. 2012] | 2012 | Tabular | simulated data; clinically relevant data (unnamed) | GLORE results identical to centrally estimated logistic regression to 10^-15 precision |
| [Jiang et al. 2013] | 2013 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Gaye et al. 2014] | 2014 | Tabular | Healthy Obese Project and Environmental Core Project (BioSHaRE-EU) data sets (10 data sets) | no number in abstract |
| [Hripcsak et al. 2015] | 2015 | EHR | no dataset (review, theory, guidance or simulation) | - |
| [Lu et al. 2015] | 2015 | Tabular | publicly available dataset (unnamed), 3 sites | no number in abstract |
| [Jochems et al. 2016] | 2016 | Tabular | Lung cancer patients from MAASTRO, Jessa, Liege, Aachen and Eindhoven (287 patients), 5 sites | AUC 0.61 (95%CI 0.51-0.70) in 5-fold cross-validation; 0.59 to 0.71 on external validation sets |
| [Johnson et al. 2016] | 2016 | EHR | MIMIC-III, 1 site | no number in abstract |
| [Mandel et al. 2016] | 2016 | EHR | no dataset (review, theory, guidance or simulation) | - |
| [Jochems et al. 2017] | 2017 | Tabular | Maastro clinic (Netherlands) cohort (559 patients); Michigan university (United States) cohort (139 patients); The Christie (United Kingdom) external validation cohort (196 patients) | AUC 0.66 on external validation set and 0.62 in 5-fold cross-validation; T and N category model AUC 0.47 on validation set. |
| [Kim et al. 2017] | 2017 | EHR | real medical datasets (multiple hospitals; not named) | no number in abstract |
| [Brisimi et al. 2018] | 2018 | EHR | patients' Electronic Health Records (unnamed) | no number in abstract |
| [Chang et al. 2018] | 2018 | Imaging | Kaggle Diabetic Retinopathy (retinal fundus photos) (35 126 color digital retinal fundus photos); Curated Breast Imaging Subset of the Digital Database for Screening Mammography (DDSM); ImageNet | no number in abstract |
| [Cho et al. 2018] | 2018 | Omics | genotype/phenotype cohorts (unnamed) (9K, 13K, and 23K individuals) | no number in abstract |

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Lee et al. 2018] | 2018 | EHR | MIMIC-III | Averaged AUC 0.9154 (balanced) and 0.8012 (imbalanced data) with k-nearest neighbor, k=3 |
| [Reps et al. 2018] | 2018 | EHR | 4 observational databases (OMOP Common Data Model) | no number in abstract |
| [Vepakomma et al. 2018] | 2018 | Multimodal | not stated | no number in abstract |
| [Choudhury et al. 2019] | 2019 | EHR | Limited MarketScan Explorys Claims-EMR Data (LCED) (over 1 million commercially insured patients); MIMIC III (over 40,000 ICU stays) | no number in abstract |
| [Huang et al. 2019] | 2019 | EHR | non-IID ICU EMRs | no number in abstract |
| [Li et al. 2019] | 2019 | Imaging | BraTS | no number in abstract |
| [Poirot et al. 2019] | 2019 | Imaging | fundus photo data set (9000 fundus photos); chest X-ray data set (156,535 chest X-rays) | Split learning performance constant for 1-50 clients vs single-center; marked difference vs non-collaborative after 2 clients ($p < 0.001$) |
| [Raisaro et al. 2019] | 2019 | Multimodal | The Cancer Genome Atlas | no number in abstract |
| [Roy et al. 2019] | 2019 | Imaging | not stated | no number in abstract |
| [Sheller et al. 2019] | 2019 | Imaging | multimodal brain scans (multi-institutional, unnamed) | Federated segmentation Dice 0.852 vs 0.862 for models trained by sharing data |
| [Suchard et al. 2019] | 2019 | EHR | global network of six administrative claims and three electronic health record databases (4·9 million patients) | Thiazide diuretics vs ACE inhibitors: acute MI HR 0·84 (0·75-0·95), heart failure hospitalisation 0·83, stroke 0·83 |
| [Beyan et al. 2020] | 2020 | Multimodal | abstract not accessible | - |
| [Chen et al. 2020b] | 2020 | Signals | not stated | 5.3% accuracy improvement for wearable activity recognition compared to traditional methods |
| [Deist et al. 2020] | 2020 | Tabular | Lung cancer patient databases of oncology departments (Personal Health Train) (23 203 patient cases), 8 sites | no number in abstract |
| [Duan et al. 2020a] | 2020 | EHR | dataset from the University of Pennsylvania Health System, 1 site | no number in abstract |
| [Duan et al. 2020b] | 2020 | EHR | 4 datasets from the Observational Health Data Sciences and Informatics (OHDSI) network; simulation study | OHDSI application: ODAC relative bias <5% for 15 of 16 log hazard ratios; meta-analysis estimates had substantially higher bias. Simulation: relative bias <0.1% vs pooled estimator. |
| [Kaissis et al. 2020] | 2020 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Li et al. 2020a] | 2020 | Imaging | ABIDE | no number in abstract |
| [Rieke et al. 2020] | 2020 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Roth et al. 2020] | 2020 | Imaging | Mammography datasets from seven clinical institutions, 7 sites | FL models 6.3% better on average than models trained on local data alone; 45.8% relative improvement in generalizability to other sites. |
| [Sheller et al. 2020] | 2020 | Imaging | multi-institutional data (10 federated institutions; not named in abstract), 10 sites | Federated learning among 10 institutions reached 99% of the model quality achieved with centralized data. |
| [Bai et al. 2021] | 2021 | Imaging | UCADI chest CT scans from hospitals in China and the UK (9,573 chest CT scans from 3,336 patients), 23 sites | FL model test sensitivity/specificity China 0.973/0.951, UK 0.730/0.942, outperforming all local models |
| [Dayan et al. 2021] | 2021 | Multimodal | data from 20 institutes across the globe (vital signs, laboratory data, chest X-rays), 20 sites | Average AUC >0.92 at 24 and 72 h; 16% average AUC improvement and 38% generalizability increase vs single-site models |
| [Dou et al. 2021] | 2021 | Imaging | multinational COVID-19 chest CT cohort (Hong Kong, Mainland China, Germany) (132 patients), 7 sites | no number in abstract |
| [Froelicher et al. 2021] | 2021 | Omics | not stated | no number in abstract |
| [Guo et al. 2021] | 2021 | Imaging | not stated | no number in abstract |
| [Kaissis et al. 2021] | 2021 | Imaging | abstract not accessible | - |
| [Liu et al. 2021] | 2021 | Imaging | retinal fundus images from public datasets, 4 sites; prostate T2-weighted MRI images from public datasets, 6 sites | Fundus segmentation: ELCFS improves over FedAvg by 2.02% Dice and 2.86 HD overall |
| [Sarma et al. 2021] | 2021 | Imaging | Private prostate MRI datasets from UCLA, SUNY Upstate Medical University and NCI (1 prostate MRI from each of 100 patients per institution), 3 sites; SPIE-AAPM-NCI PROSTATEx (challenge dataset) (343 axial T2W images) | FL model overall mean Dice 0.889, significantly superior to every single-institution model ($P < .001$) |
| [Vaid et al. 2021] | 2021 | EHR | Mount Sinai Health System EHR data, 5 sites | no number in abstract |

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Warnat-Herresthal et al. 2021] | 2021 | Multimodal | blood transcriptomes from 127 clinical studies (more than 16,400 blood transcriptomes); chest X-ray images (more than 95,000 chest X-ray images) | no number in abstract |
| [Xu et al. 2021] | 2021 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Ziller et al. 2021] | 2021 | Imaging | paediatric pneumonia dataset; Medical Segmentation Decathlon Liver dataset | no number in abstract |
| [Adnan et al. 2022] | 2022 | Imaging | The Cancer Genome Atlas (TCGA) | no number in abstract |
| [Jiang et al. 2022] | 2022 | Imaging | Camelyon17 (450,000 histology images), 5 sites; MoNuSAC2020; MoNuSAC2018; multi-site prostate segmentation dataset (6 data sources from 3 public datasets) | no number in abstract |
| [Lu et al. 2022] | 2022 | Imaging | histology whole slide images (unnamed) (thousands of histology whole slide images) | no number in abstract |
| [Nguyen et al. 2022] | 2022 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Ogier du Terrail et al. 2022] | 2022 | Multimodal | FLamby (7 healthcare datasets with natural splits) (7 datasets) | no number in abstract |
| [Pati et al. 2022] | 2022 | Imaging | glioblastoma data from 71 sites across 6 continents (n = 6,314), 71 sites | 33% delineation improvement for surgically targetable tumor and 23% for complete tumor extent over a publicly trained model. |
| [Saldanha et al. 2022] | 2022 | Imaging | Epi700 (661 patients); DACHS (2,448 patients); TCGA (632 patients); QUASAR; YCR BCIP | MSI in QUASAR: swarm AUROCs 0.8001 and 0.8151 vs 0.7884 for Epi700 single-cohort model |
| [Tedeschini et al. 2022] | 2022 | Imaging | representative clinical datasets obtained from the daily clinical workflow | no number in abstract |
| [Xu et al. 2022] | 2022 | Imaging | not stated | no number in abstract |
| [Kalra et al. 2023] | 2023 | Imaging | popular image datasets; gigapixel histology whole slide images (cancer diagnostic problem) | no number in abstract |
| [Ogier du Terrail et al. 2023] | 2023 | Imaging | multicentric TNBC whole-slide image and clinical data | no number in abstract |
| [Yan et al. 2023] | 2023 | Imaging | simulated and real-world medical imaging non-IID federated datasets (retinal, dermatology, chest X-ray) | Under severe heterogeneity, test accuracy improved 5.06%, 1.53%, 4.58% (retinal, dermatology, chest X-ray) vs supervised baseline with ImageNet pre-training. |

**P2 Proxy (47 works)**

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Choi et al. 2016] | 2016 | EHR | longitudinal time stamped EHR data (unnamed) (260K patients over 8 years) | Differential diagnosis up to 79% recall@30, significantly higher than several baselines |
| [Gulshan et al. 2016] | 2016 | Imaging | retrospective development data set (128 175 retinal images); EyePACS-1 (9963 images from 4997 patients); Messidor-2 (1748 images from 874 patients) | AUC 0.991 (EyePACS-1) and 0.990 (Messidor-2) for referable diabetic retinopathy |
| [Shin et al. 2016] | 2016 | Imaging | not stated | no number in abstract |
| [Tajbakhsh et al. 2016] | 2016 | Imaging | not stated | no number in abstract |
| [Esteva et al. 2017] | 2017 | Imaging | clinical skin lesion images (129,450 clinical images) | no number in abstract |
| [Kamnitsas et al. 2017] | 2017 | Imaging | two MR databases of subjects with traumatic brain injuries | no number in abstract |
| [Alsentzer et al. 2019] | 2019 | Text | MIMIC-III v1.4 (approximately 2 million notes); MedNLI; i2b2 2006; i2b2 2010; i2b2 2012; i2b2 2014 | MedNLI accuracy 82.7% vs prior state of the art 73.5% |
| [Chen et al. 2019] | 2019 | Imaging | 3DSeg-8 (aggregated from several medical challenges); LIDC; LiTS challenge | 2x faster convergence vs Kinetics pretraining, 10x vs scratch; accuracy gains 3% to 20%; 94.6% Dice on LiTS |
| [Gurovich et al. 2019] | 2019 | Imaging | community-driven phenotyping platform facial images (over 17,000 images) | DeepGestalt 91% top-10 accuracy identifying the correct syndrome on 502 images |
| [Prabhu et al. 2019] | 2019 | Imaging | realistic dataset of dermatological conditions | no number in abstract |
| [Raghu et al. 2019] | 2019 | Imaging | not stated | no number in abstract |
| [Zhang et al. 2019] | 2019 | EHR | real patient EHR repository from Oregon Health & Science University, 1 site | no number in abstract |
| [Lee et al. 2020] | 2020 | Text | PubMed abstracts (4.5B words); PubMed Central (PMC) full-text articles (13.5B words) | Vs previous SOTA: NER +0.62% F1, relation extraction +2.80% F1, QA +12.24% MRR |

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Li et al. 2020b] | 2020 | EHR | Clinical Practice Research Datalink (CPRD) (nearly 1.6 million individuals), 674 sites | 8.0-13.2% improvement in average precision over state-of-the-art deep EHR models |
| [Mahajan et al. 2020] | 2020 | Imaging | abstract not accessible | - |
| [Roy et al. 2020] | 2020 | Imaging | Visceral Dataset | no number in abstract |
| [Azizi et al. 2021] | 2021 | Imaging | dermatology digital camera images; chest X-ray images (multi-label) | +6.7% top-1 accuracy (dermatology) and +1.1% mean AUC (chest X-ray) over supervised ImageNet-pretrained baselines. |
| [Huang et al. 2021b] | 2021 | Imaging | not stated | no number in abstract |
| [Rasmy et al. 2021] | 2021 | EHR | Cerner Health Facts (28,490,650 patients); Truven Health MarketScan | AUC improved by 1.21-6.14% in two disease prediction tasks; >20% with small fine-tuning sets |
| [Sowrirajan et al. 2021] | 2021 | Imaging | CheXpert (224k chest X-rays collected from 65k patients); Shenzhen Hospital X-ray set (662 X-ray images) | Pleural effusion, 0.1% labels: linear model AUC 0.776 with MoCo-CXR vs 0.683 ImageNet-pretrained (+0.096) |
| [Steinberg et al. 2021] | 2021 | EHR | not stated | 3.5% mean AUROC improvement over standard baselines on five tasks; 19% average improvement with few training records |
| [Zhou et al. 2021] | 2021 | Imaging | not stated | no number in abstract |
| [Agrawal et al. 2022] | 2022 | Text | CASI dataset (re-annotated) | no number in abstract |
| [Guan and Liu 2022] | 2022 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Hsieh et al. 2022] | 2022 | Imaging | patient photographs (Clinical Face Phenotype Space) (17,560 patients with 1,115 rare disorders) | no number in abstract |
| [Mei et al. 2022] | 2022 | Imaging | RadImageNet (1.35 million annotated medical images in 131 872 patients) | no number in abstract |
| [Tiu et al. 2022] | 2022 | Imaging | not stated | no number in abstract |
| [Wang et al. 2022] | 2022 | Imaging | not stated | With only 20K pre-training data, MedCLIP outperforms the state-of-the-art method using ≈200K data |
| [Zhang et al. 2022] | 2022 | Imaging | not stated | Needs only 10% as much labeled data as ImageNet-initialized counterpart for better or comparable performance on 4 classification tasks. |
| [Banerjee et al. 2023] | 2023 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Hegselmann et al. 2023] | 2023 | Tabular | Bank (45,211 rows); Blood (748 rows); California (20,640 rows); Car (1,728 rows); Credit-g (1,000 rows); Income (48,842 rows); Jungle (44,819 rows); Diabetes (Kaggle) (768 rows); Heart (Kaggle) (918 rows); de-identified health claims dataset from a U.S. health insurer | Zero-shot AUC on health claims tasks from 0.67 (Surgery) to 0.71 (LoH) |
| [Hollmann et al. 2023] | 2023 | Tabular | OpenML-CC18 suite (18 datasets) (up to 1 000 training data points each); additional small numerical datasets from OpenML (67 datasets) | Outperforms boosted trees, on par with state-of-the-art AutoML with up to 230x speedup (5 700x on GPU) |
| [Jiang et al. 2023] | 2023 | Text | clinical notes from the authors' health system | NYUTron AUC 78.7-94.9%, improving AUC by 5.36-14.7% over traditional models |
| [Moor et al. 2023] | 2023 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Nori et al. 2023] | 2023 | Text | USMLE official practice materials; MultiMedQA | GPT-4 exceeds USMLE passing score by over 20 points; outperforms GPT-3.5 and Med-PaLM. |
| [Singhal et al. 2023] | 2023 | Text | MultiMedQA; HealthSearchQA; MedQA; MedMCQA; PubMedQA; MMLU clinical topics | Flan-PaLM 67.6% accuracy on MedQA, surpassing prior state of the art by more than 17%. |
| [Theodoris et al. 2023] | 2023 | Omics | large-scale corpus of single-cell transcriptomes (unnamed) (about 30 million single-cell transcriptomes) | no number in abstract |
| [Wornow et al. 2023] | 2023 | EHR | no dataset (review, theory, guidance or simulation) | - |
| [Zhou et al. 2023] | 2023 | Imaging | unlabelled retinal images (RETFound pretraining) (1.6 million unlabelled retinal images) | no number in abstract |
| [Chen et al. 2024] | 2024 | Imaging | diagnostic H&E-stained WSIs across 20 major tissue types (pretraining) (more than 100 million images from over 100,000 WSIs) | no number in abstract |
| [Cui et al. 2024] | 2024 | Omics | single-cell sequencing data repository (over 33 million cells) | no number in abstract |

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Lu et al. 2024] | 2024 | Imaging | histopathology image-caption pairs from diverse sources (CONCH pretraining) (over 1.17 million image-caption pairs) | no number in abstract |
| [Ma et al. 2024] | 2024 | Imaging | curated dataset from publicly available medical image segmentation datasets (1,570,263 image-mask pairs) | no number in abstract |
| [Pai et al. 2024] | 2024 | Imaging | DeepLesion (11,467 radiographic lesions used for pretraining; dataset comprises 32,735 lesions from 4,427 unique patients); LUNA16 (507 lung nodules); HarvardRT (n = 291); LUNG1 (NSCLC-Radiomics) (n = 420); RADIO (NSCLC-Radiogenomics) (n = 133) | LUNG1 2-year survival: Foundation (features) AUC 0.638 (95% CI 0.584-0.692), exceeding all baselines |
| [Vorontsov et al. 2024] | 2024 | Imaging | not stated | Pan-cancer detection specimen-level AUC 0.95 across nine common and seven rare cancers |
| [Xu et al. 2024] | 2024 | Imaging | Providence health network whole slides (171,189 whole slides (1.3 billion tiles; more than 30,000 patients)), 28 sites; TCGA | State-of-the-art on 25 of 26 tasks; significant improvement over second-best method on 18 tasks |
| [Hollmann et al. 2025] | 2025 | Tabular | synthetic datasets (pretraining) (millions of synthetic datasets) | In 2.8 s, TabPFN outperforms an ensemble of strongest baselines tuned for 4 h (classification, datasets up to 10,000 samples). |

**P3 Prior (15 works)**

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Steyerberg et al. 2000] | 2000 | Tabular | small data sets of patients with acute myocardial infarction | no number in abstract |
| [Choi et al. 2017a] | 2017 | EHR | electronic health records (unnamed) | Vs basic RNN: 10% higher accuracy for rare diseases; 3% higher heart-failure AUC |
| [Rotmensch et al. 2017] | 2017 | EHR | de-identified patient records (273,174 de-identified patient records) | Noisy OR model precision 0.85 at recall 0.6 in clinical evaluation; outperforms logistic regression and naive Bayes ($p < 0.01$) |
| [Ma et al. 2018] | 2018 | EHR | abstract not accessible | - |
| [Shang et al. 2019] | 2019 | EHR | real EHR data | Outperformed all baselines in all effectiveness measures; 3.60% DDI rate reduction from existing EHR data |
| [Subbaswamy et al. 2019] | 2019 | EHR | not stated | no number in abstract |
| [Castro et al. 2020] | 2020 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Kissas et al. 2020] | 2020 | Imaging | in-vivo 4D flow MRI measurements near the aorta/carotid bifurcation (a healthy human subject) | no number in abstract |
| [Lagergren et al. 2020] | 2020 | Tabular | wound healing assays with varying initial cell densities | no number in abstract |
| [Sahli Costabal et al. 2020] | 2020 | Signals | abstract not accessible | - |
| [Van Calster et al. 2020] | 2020 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Köhler et al. 2021] | 2021 | Text | no dataset (review, theory, guidance or simulation) | - |
| [Lu et al. 2021a] | 2021 | Tabular | PK data from two treatment regimens of trastuzumab emtansine | no number in abstract |
| [Riley et al. 2021] | 2021 | Tabular | not stated | no number in abstract |
| [Zhang et al. 2021] | 2021 | Multimodal | eICU collaborative research database V2.0 (30,680 patients (study cohort)), over 200 sites; MIMIC-CXR; CheXpert; Chest-Xray8; PadChest | no number in abstract |

**P4 Unlabeled (18 works)**

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Agarwal et al. 2016] | 2016 | EHR | not stated | Precision/accuracy 0.90/0.89 (T2DM) and 0.86/0.89 (MI) vs rule-based 0.96/0.92 and 0.84/0.87 |
| [Halpern et al. 2016] | 2016 | EHR | retrospective EMR data on emergency department patients | Triage-time AUCs: infection 0.89, cancer 0.88, septic shock 0.93, pneumonia 0.90; disposition-time AUCs up to 0.97 (septic shock, pneumonia) |
| [Bai et al. 2017] | 2017 | Imaging | abstract not accessible | - |

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Wang et al. 2017] | 2017 | Imaging | ChestX-ray8 (108,948 frontal-view X-ray images of 32,717 unique patients) | no number in abstract |
| [Ilse et al. 2018] | 2018 | Imaging | MNIST-bags (from MNIST); Breast cancer histopathology dataset (Gelasca et al., 2008) (58 H&E images); Colon cancer histopathology dataset (Sirinukunwattana et al., 2016) (100 H&E images) | no number in abstract |
| [Campanella et al. 2019] | 2019 | Imaging | whole slide image dataset (44,732 whole slide images from 15,187 patients) | AUC above 0.98 for prostate cancer, basal cell carcinoma and breast cancer metastases |
| [Cheplygina et al. 2019] | 2019 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Fries et al. 2019] | 2019 | Imaging | UK Biobank (up to 4,000 unlabeled cardiac MRI sequences) | Model identifies individuals with a 1.8-fold increase in risk of a major adverse cardiac event; outperforms supervised hand-labeled model |
| [Irvin et al. 2019] | 2019 | Imaging | CheXpert (224,316 chest radiographs of 65,240 patients) | no number in abstract |
| [Wang et al. 2019] | 2019 | Text | Mayo Clinic smoking status classification (institutional), 1 site; Mayo Clinic proximal femur (hip) fracture classification (institutional), 1 site; i2b2 2006 smoking status classification shared task | CNN best: F1 0.92 (Mayo smoking status) and 0.97 (fracture classification). |
| [Yu et al. 2019] | 2019 | Imaging | MICCAI 2018 Atrial Segmentation Challenge dataset (100 3D gadolinium-enhanced MR imaging scans) | With 16 labeled scans, unlabeled data improves Dice by 2.85% and Jaccard 4.15% over labeled-only Bayesian V-Net |
| [Chaitanya et al. 2020] | 2020 | Imaging | ACDC (100 3D short-axis cardiac cine-MRIs); Prostate (MICCAI 2018 medical segmentation decathlon) (48 3D T2-weighted MRIs); MMWHS (20 3D cardiac MRIs) | Within 8% of benchmark using only two labeled MRI volumes (4% of ACDC training data) |
| [Ouyang et al. 2020] | 2020 | Imaging | not stated | no number in abstract |
| [Tajbakhsh et al. 2020] | 2020 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Taleb et al. 2020] | 2020 | Imaging | 3D MRI (Brain Tumor Segmentation); 3D CT (Pancreas Tumor Segmentation); 2D Fundus images (Diabetic Retinopathy Detection) | no number in abstract |
| [Li et al. 2021c] | 2021 | Imaging | International Skin Imaging Collaboration (ISIC) 2017; Retinal Fundus Glaucoma Challenge (REFUGE); Liver Tumor Segmentation Challenge (LiTS) | no number in abstract |
| [Lu et al. 2021b] | 2021 | Imaging | not stated | no number in abstract |
| [Krishnan et al. 2022] | 2022 | Multimodal | no dataset (review, theory, guidance or simulation) | - |

**P5 Procurement (12 works)**

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Peduzzi et al. 1996] | 1996 | Tabular | cardiac trial data (673 patients) | EPV of 10 or greater: no major problems; EPV below 10: biased coefficients and poor variance/coverage |
| [Figueroa et al. 2012] | 2012 | Text | learning curves from clinical text and waveform classification tasks | 80 to 560 annotated samples needed for mean average and RMSE below 0.01; weighted fitting outperformed un-weighted baseline |
| [Cho et al. 2015] | 2015 | Imaging | Massachusetts General Hospital (MGH) PACS CT images (6000 test CT images; training sizes 5-200), 1 site | no number in abstract |
| [Yang et al. 2017] | 2017 | Imaging | 2015 MICCAI Gland Challenge dataset; lymph node ultrasound image segmentation dataset | State-of-the-art segmentation performance achieved using only 50% of training data. |
| [Smailagic et al. 2018] | 2018 | Imaging | Messidor (1200 eye fundus images); Breast Cancer Diagnosis Dataset (ICIAR 2018 Grand Challenge) (400 images); Skin Cancer dataset (900 images) | 80% DR detection accuracy with 425 labeled images; 32% fewer labels than uncertainty sampling, 40% fewer than random |
| [Balki et al. 2019] | 2019 | Imaging | no dataset (review, theory, guidance or simulation) | 22 of 167 articles included; only 4 discussed sample-size determination methodologies |
| [Riley et al. 2019a] | 2019 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Riley et al. 2019b] | 2019 | Tabular | no dataset (review, theory, guidance or simulation) | - |

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [van Smeden et al. 2019] | 2019 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Riley et al. 2020] | 2020 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Budd et al. 2021] | 2021 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Nath et al. 2021] | 2021 | Imaging | MRI scans of the hippocampus; CT scans of pancreas and tumors | Full accuracy using only 22.69% and 48.85% of available data for each dataset |

**P6 Re-expression (26 works)**

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Chawla et al. 2002] | 2002 | Tabular | not stated | no number in abstract |
| [Choi et al. 2017b] | 2017 | EHR | not stated | no number in abstract |
| [Bowles et al. 2018] | 2018 | Imaging | not stated | Dice Similarity Coefficient improvements of 1 to 5 percentage points from adding GAN-derived synthetic training data. |
| [Fernández et al. 2018] | 2018 | Tabular | no dataset (review, theory, guidance or simulation) | - |
| [Frid-Adar et al. 2018] | 2018 | Imaging | CT images of liver lesions (limited dataset) (182 liver lesions) | Sensitivity/specificity 85.7%/92.4% with synthetic augmentation vs 78.6%/88.4% with classic augmentation |
| [Shin et al. 2018] | 2018 | Imaging | Alzheimer's Disease Neuroimaging Initiative (ADNI) (3,416 pairs of T1-weighted images); BRATS 2015 training data set (264 studies) | no number in abstract |
| [Walonoski et al. 2018] | 2018 | EHR | Synthea synthetic patient records (One million synthetic patient records) | no number in abstract |
| [Baowaly et al. 2019] | 2019 | EHR | MIMIC-III; National Health Insurance Research Database (NHIRD), Taiwan | no number in abstract |
| [Beaulieu-Jones et al. 2019] | 2019 | Tabular | SPRINT trial (Systolic Blood Pressure Trial) | no number in abstract |
| [Jordon et al. 2019] | 2019 | Tabular | abstract not accessible | - |
| [Xu et al. 2019] | 2019 | Tabular | adult (UCI); census (UCI); covertype (UCI); intrusion (UCI); news (UCI); credit (Kaggle); MNIST28 / MNIST12 (from MNIST); 7 simulated datasets | CTGAN surpasses Bayesian networks on at least 87.5% of benchmark datasets |
| [Yi et al. 2019] | 2019 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Zhao et al. 2019] | 2019 | Imaging | not stated | no number in abstract |
| [Goncalves et al. 2020] | 2020 | EHR | Surveillance Epidemiology and End Results (SEER) cancer registry (over 360,000 individual cases) | no number in abstract |
| [Yoon et al. 2020] | 2020 | EHR | four independent datasets (4 datasets) | no number in abstract |
| [Zhang et al. 2020] | 2020 | EHR | billing codes from EMRs at Vanderbilt University Medical Center (over 1 million EMRs), 1 site | no number in abstract |
| [Chaitanya et al. 2021] | 2021 | Imaging | cardiac; prostate; pancreas | no number in abstract |
| [Chen et al. 2021a] | 2021 | Multimodal | abstract not accessible | - |
| [Chlap et al. 2021] | 2021 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Stadler et al. 2022] | 2022 | Tabular | not stated | no number in abstract |
| [van den Goorbergh et al. 2022] | 2022 | Tabular | ovarian cancer diagnosis case study data | no number in abstract |
| [Borisov et al. 2023] | 2023 | Tabular | numerous real-world and synthetic data sets (not named) | no number in abstract |
| [Kazerouni et al. 2023] | 2023 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Khader et al. 2023] | 2023 | Imaging | ADNI (brain MRI) (998 studies); LIDC (chest CT) (1010 studies); DUKE (breast MRI) (1844 studies); MRNet (knee MRI) (1250 studies) | Breast segmentation Dice 0.95 with synthetic data vs 0.91 without |
| [Kotelnikov et al. 2023] | 2023 | Tabular | not stated | no number in abstract |
| [Ktena et al. 2024] | 2024 | Imaging | histopathology images; chest X-ray images; dermatology images | no number in abstract |

**No remedy (documents a problem) (26 works)**

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [El Emam et al. 2011] | 2011 | Tabular | no dataset (review, theory, guidance or simulation) | Pooled re-identified proportion 0.26 (95% CI 0.046-0.478) overall and 0.34 for health data, across 14 attacks. |
| [Varoquaux 2018] | 2018 | Imaging | not stated | Sample sizes of many neuroimaging studies lead to large cross-validation error bars, e.g. ±10% for 100 samples. |
| [Zech et al. 2018] | 2018 | Imaging | National Institutes of Health Clinical Center (NIH) (112,120 radiographs from 30,805 patients), 1 site; Mount Sinai Hospital (MSH) (42,396 radiographs from 12,904 patients), 1 site; Indiana University Network for Patient Care (IU) (3,807 radiographs from 3,683 patients), 1 site | Sorting by hospital system alone achieved AUC 0.861 on joint MSH-NIH data; better internal than external in 3/5 comparisons |
| [Brasil et al. 2019] | 2019 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Christodoulou et al. 2019] | 2019 | Tabular | no dataset (review, theory, guidance or simulation) | Low-risk-of-bias comparisons (145): logit(AUC) difference LR vs ML 0.00 (95% CI -0.18 to 0.18) |
| [Kim et al. 2019] | 2019 | Imaging | no dataset (review, theory, guidance or simulation) | Of 516 eligible studies, only 6% (31 studies) performed external validation |
| [Liu et al. 2019] | 2019 | Imaging | no dataset (review, theory, guidance or simulation) | Pooled sensitivity 87.0% (DL) vs 86.4% (clinicians); pooled specificity 92.5% vs 90.5%, in 14 studies. |
| [Obermeyer et al. 2019] | 2019 | EHR | not stated | Remedying the bias would increase Black patients receiving additional help from 17.7% to 46.5%. |
| [Rocher et al. 2019] | 2019 | Tabular | 210 populations (unnamed) (210 populations) | AUC 0.84-0.97 for predicting individual uniqueness; 99.98% of Americans re-identifiable with 15 attributes |
| [Schwarz et al. 2019] | 2019 | Imaging | deidentified cranial MRI scans | no number in abstract |
| [Nagendran et al. 2020] | 2020 | Imaging | no dataset (review, theory, guidance or simulation) | High overall risk of bias in 58 of 81 non-randomised studies; data and code unavailable in 95% and 93% |
| [Nguengang Wakap et al. 2020] | 2020 | Tabular | Orphanet database (6172 unique rare diseases) | Population prevalence of rare diseases estimated at 3.5-5.9%, i.e. 263-446 million persons globally |
| [Oakden-Rayner et al. 2020] | 2020 | Imaging | CIFAR-100; multiple real-world medical imaging datasets | Hidden stratification can cause relative performance differences of over 20% on clinically important subsets. |
| [Schaefer et al. 2020] | 2020 | Multimodal | no dataset (review, theory, guidance or simulation) | 211 studies from 32 countries on 74 rare diseases; only 11.8% evaluated on an external data set. |
| [Willemink et al. 2020] | 2020 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Wynants et al. 2020] | 2020 | Multimodal | no dataset (review, theory, guidance or simulation) | 169 studies, 232 models; all at high or unclear risk of bias; C index 0.54 to >0.99 |
| [Chen et al. 2021b] | 2021 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Finlayson et al. 2021] | 2021 | Multimodal | no dataset (review, theory, guidance or simulation) | - |
| [Koh et al. 2021] | 2021 | Multimodal | WILDS benchmark (10 datasets) | no number in abstract |
| [Roberts et al. 2021] | 2021 | Imaging | no dataset (review, theory, guidance or simulation) | 61 studies included (of 2,212 identified); none of the models identified are of potential clinical use. |
| [Wong et al. 2021] | 2021 | EHR | Michigan Medicine (University of Michigan) hospitalized patients (27,697 patients with 38,455 hospitalizations), 1 site | Epic Sepsis Model hospitalization-level AUROC 0.63 (95% CI 0.62-0.64) |
| [Gichoya et al. 2022] | 2022 | Imaging | Emory CXR; Emory Chest CT; Emory Cervical Spine; Emory Mammogram; MIMIC-CXR; CheXpert; National Lung Cancer Screening Trial; RSNA Pulmonary Embolism CT; Digital Hand Atlas | Race prediction AUC 0·91-0·99 (x-ray), 0·87-0·96 (CT chest), 0·81 (mammography) under external validation |
| [Packhäuser et al. 2022] | 2022 | Imaging | ChestX-ray14 (112,120 frontal-view chest X-ray images from 30,805 unique patients); CheXpert; COVID-19 Image Data Collection | Same-patient verification AUC 0.9940, accuracy 95.55% on ChestX-ray14; external datasets AUC up to 0.9870 |
| [Varoquaux and Cheplygina 2022] | 2022 | Imaging | no dataset (review, theory, guidance or simulation) | - |
| [Yu et al. 2022] | 2022 | Imaging | no dataset (review, theory, guidance or simulation) | 81% (70 of 86) algorithms showed decreased external vs internal performance; 24% a substantial decrease (>=0.10). |

| Work | Year | Modality | Data (size) | Reported result |
|---|---|---|---|---|
| [Hatamizadeh et al. 2023] | 2023 | Imaging | not stated | no number in abstract |

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